

User Manual for the Turn by Turn Analysis GUI

version 0.1

March 23, 2025

The Turn-by-turn analysis GUI is a tool designed to provide insights into the beam dynamics during the first turns after an injection. Please note that this tool is intended for expert use only. You can launch the GUI by executing the following command:

```
bash /home/als/physbase/users/thellert/automated_startup/GUI/start_TxT_gui.sh
```

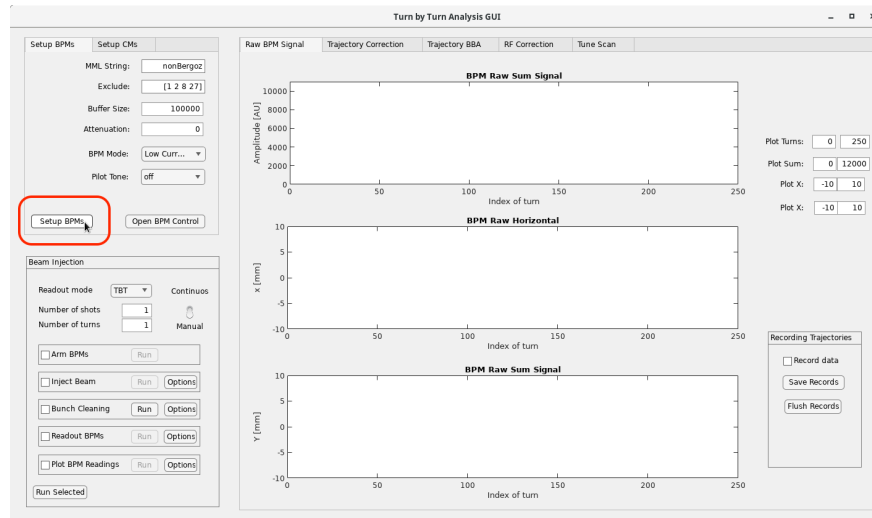
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1 Set Up BPMs

Instruction: set up the BPM structure in the GUI by pressing the **Setup BPMs** button.

Note: This step ensures that the correct BPM settings are loaded, as seen in the optional fields above. It is recommended to not change those unless you know what you do.

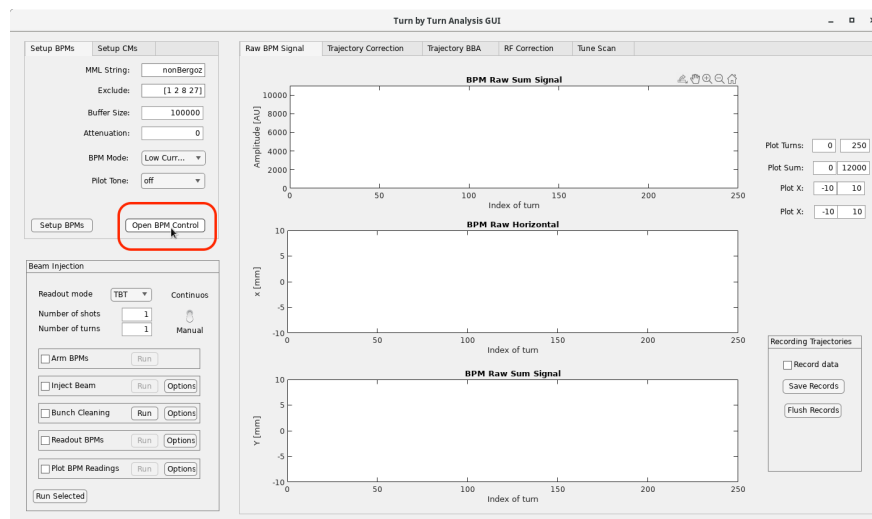


Step 1: Loading the Correct BPM Settings

2 Open BPM Control Panel

Instruction: Open the BPM control panel by pressing the **Open BPM control** button.

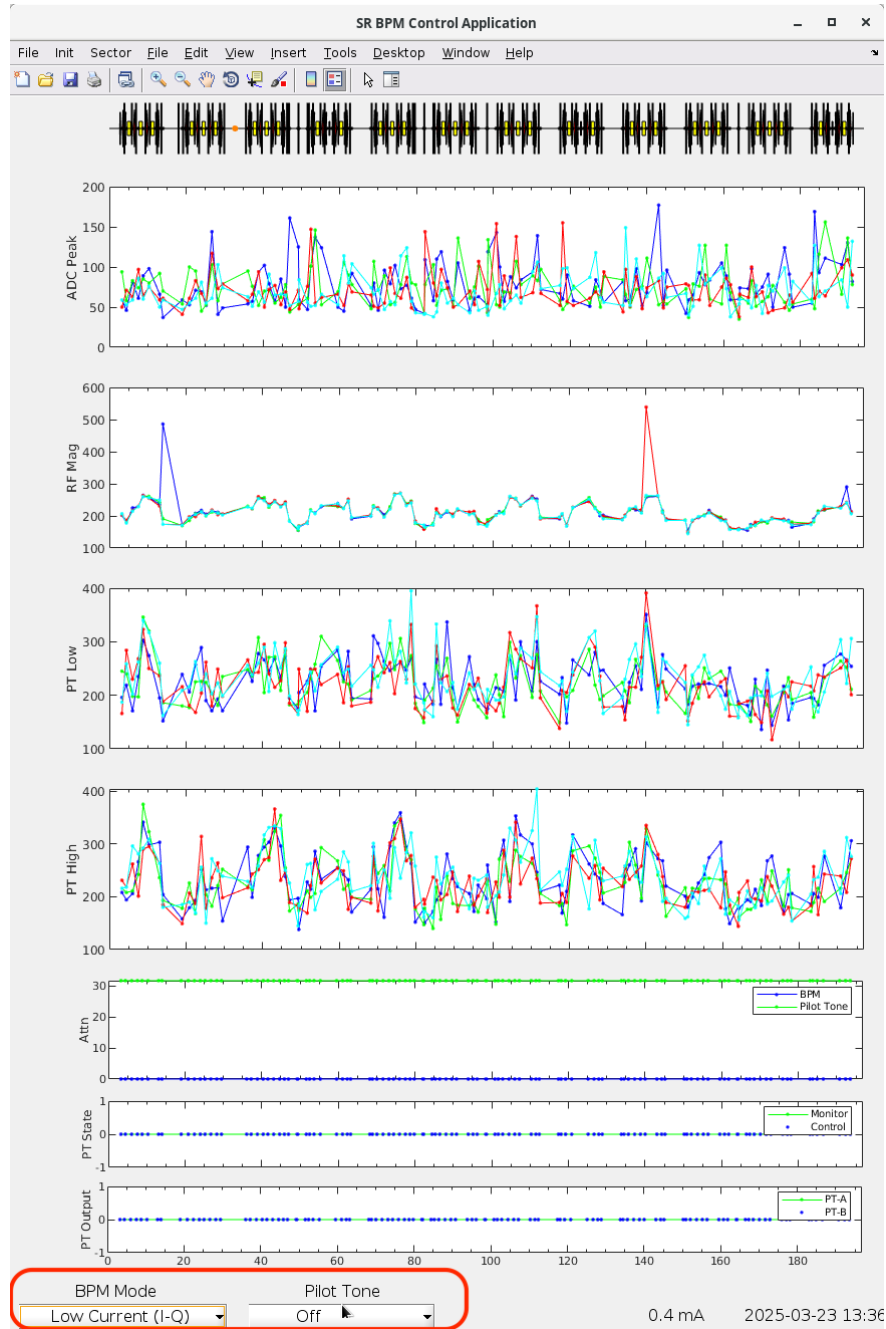
Note: For some reason, the pilot tone and bunch current mode can often not be changed from this MATLAB session. Thus, you have to open the BPM Control panel and do it there manually.



Step 2: Opening the BPM Control Panel

3 BPM Control Panel Settings

Instruction: This is the BPM control panel. Make sure the pilot tone is switched off and the beam current is set to low.



Step 3: BPM Panel Settings: Pilot Tone Off and Low Beam Current

Instruction: Now select the steps that should be included in your injection chain. Typically this means:

- If the RF is switched off, bunch cleaning is not required; otherwise, bunch cleaning can be included (note that bunch cleaning requires fairly well-corrected tunes).

Turn by Turn Analysis GUI

Setup BPMs

MML String:

Exclude:

Buffer Size:

Attenuation:

BPM Mode:

Pilot Tone:

Raw BPM Signal Trajectory Correction Trajectory BBA RF Correction Tune Scan

BPM Raw Sum Signal

Plot Turns:

Plot Sum:

Plot X:

Plot Y:

BPM Raw Horizontal

BPM Raw Vertical

Beam Injection

Readout mode:

Number of shots:

Number of turns:

☒ Arm BPMs

☒ Inject Beam

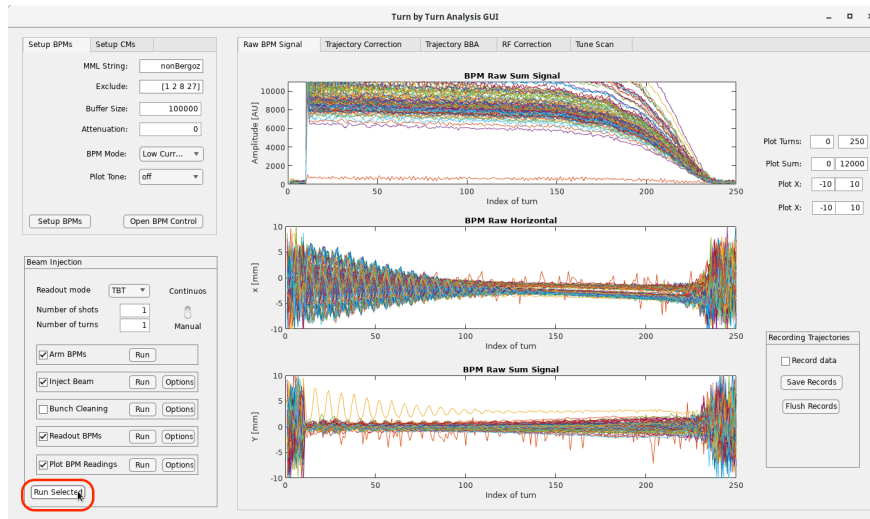
☐ Bunch Cleaning

☒ Readout BPMs

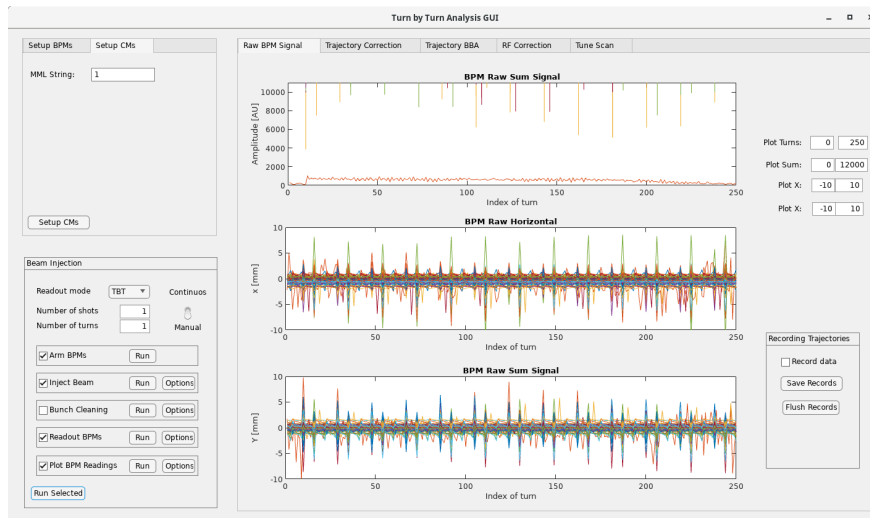
☒ Plot BPM Readings

5 Run Selected and Observe Injection Results

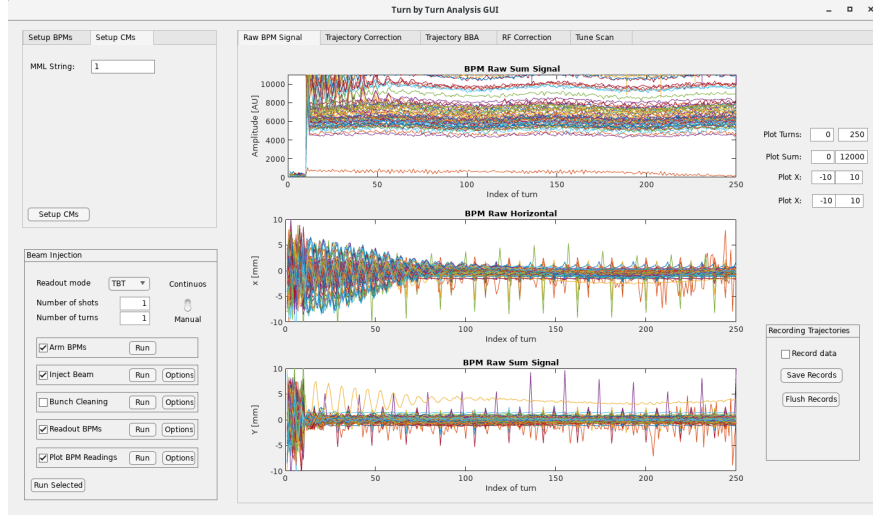
- Initially, there is no injection; the sum signal is zero, and the X and Y signals show some noisy activity with high amplitudes.
- Then, the injected beam appears on all BPM sum signals.
- Without the RF cavity, the beam dies off at about 200 turns, revealing de-coherence in the horizontal trajectories.



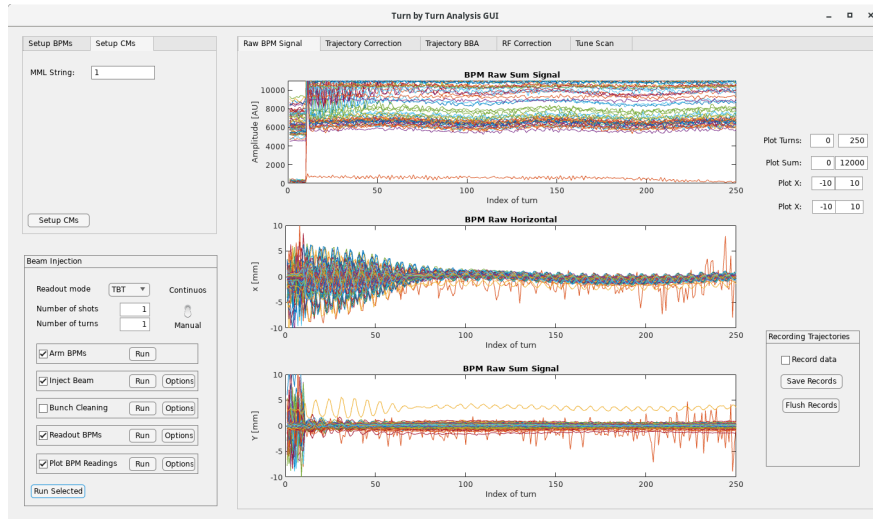
Step 5: This is what a typical injection would look like.



Step 5: If you see very high sum signals and the spikes in the horizontal and vertical planes, the pilot tone is still switched on



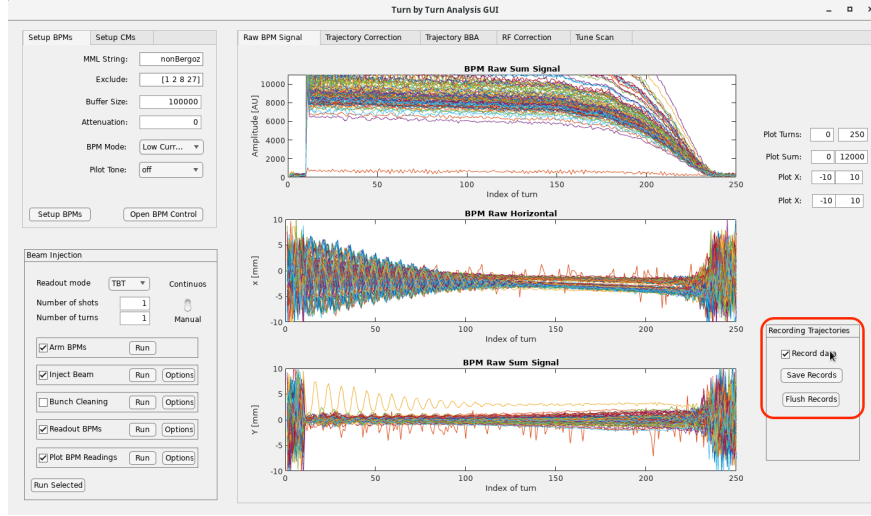
Step 5: This is how an injection with RF on looks like. The beam does not die off.



Step 5: This is how it looks when you inject and there's beam in the machine. You can see that the sum signal is not zero at the beginning.

6 Record Injection Shots

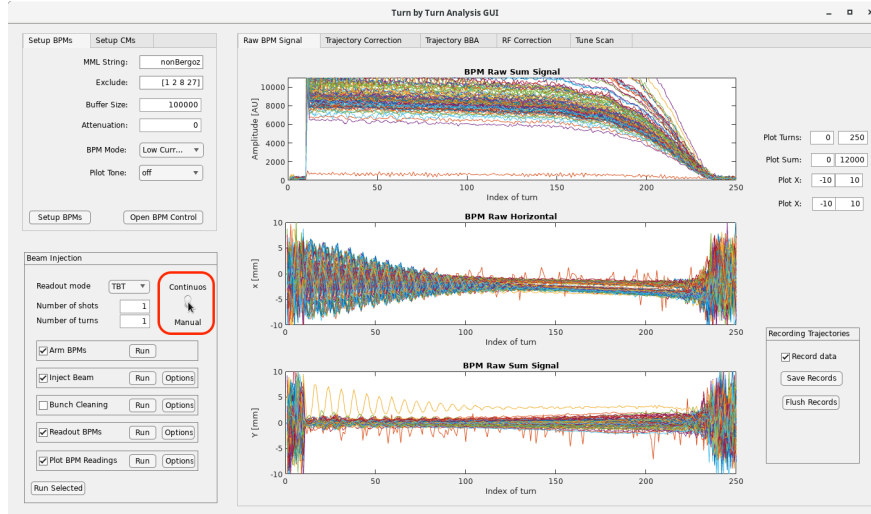
Instruction: You can choose to record every injection shot and save the records if desired for offline analysis. To remove all stored trajectories from the local memory, click **flush records**.



Step 6: Recording and Managing Injection Shots

7 Continuous Mode Injection

Instruction: You can also inject in continuous mode, meaning that the selected injection chain will run in a loop.



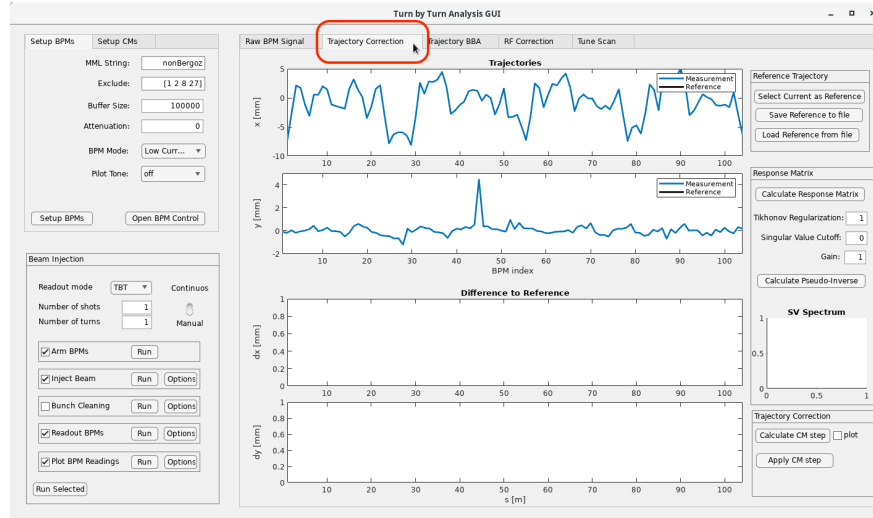
Step 7: Continuous Mode Operation for Injection

8 One Turn Trajectory

Instruction: Now you can look at the trajectory panel at the top. This is how a one-turn trajectory could look.

Note: During startup, when beam transmission is very low, accurately determining beam loss can be challenging. The app does not define beam loss explicitly; instead, it uses a cutoff in the sum signal to infer loss. If the threshold is set too high, you might observe an early indication of

beam loss. Conversely, if the threshold is too low, the displayed horizontal and vertical trajectories may simply represent noise.

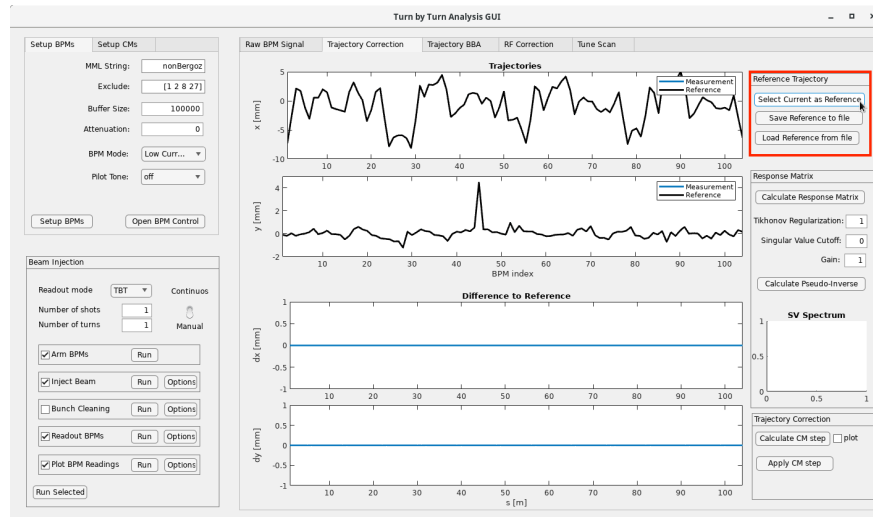


Step 8: Viewing a One-Turn Trajectory

9 Set Reference Trajectory

Instruction: In this example, we select the current trajectory to be used as the reference.

Note: You can also choose to save the reference or load a reference from a file which is very useful for startups.



Step 9: Setting and Managing the Reference Trajectory

10 Shot-to-Shot Variation

Instruction: Now let's inject another time and see what the shot-to-shot variation looks like.



Step 10: Examining Shot-to-Shot Variation in Injection

11 Trajectory Correction Demonstration

Instruction: In order to demonstrate trajectory correction, for this example, we excited a vertical corrector magnet around BPM 50 to induce a noticeable trajectory variation in the vertical plane.



Step 11: Exciting a Vertical Corrector Magnet to Induce Trajectory Variation

12 Set Up Corrector Magnets

Instruction: For trajectory correction to work, we need to set up the corrector magnets. This can be done by pushing the **Setup CMs** button in the corresponding panel.

Note: The set of corrector magnets are selected based on the MATLAB middle layer string (see MML documentation for details).



Step 12: Setting Up Corrector Magnets via the Setup CMs Button

13 Calculate Response Matrix

Instruction: Now that we have our corrector magnets selected, we can calculate the response matrix.



Step 13: Calculation of the Response Matrix

14 Calculate Pseudo-Inverse Response Matrix

Instruction: Calculate the pseudo-inverse response matrix given a set of regularization parameters.

Note: By default, Tikhonov regularization is used, but you can also switch to a classic singular value cutoff by defining the number of cutoff singular values. In that case, set Tikhonov regular-

ization to zero. The gain is multiplied to the inverse response matrix and is effectively a gain in the applied correction step.



Step 14: Pseudo-Inverse Response Matrix Calculation with Regularization

15 Calculate Corrector Magnet Step

Instruction: Now calculate the corrector magnet step.

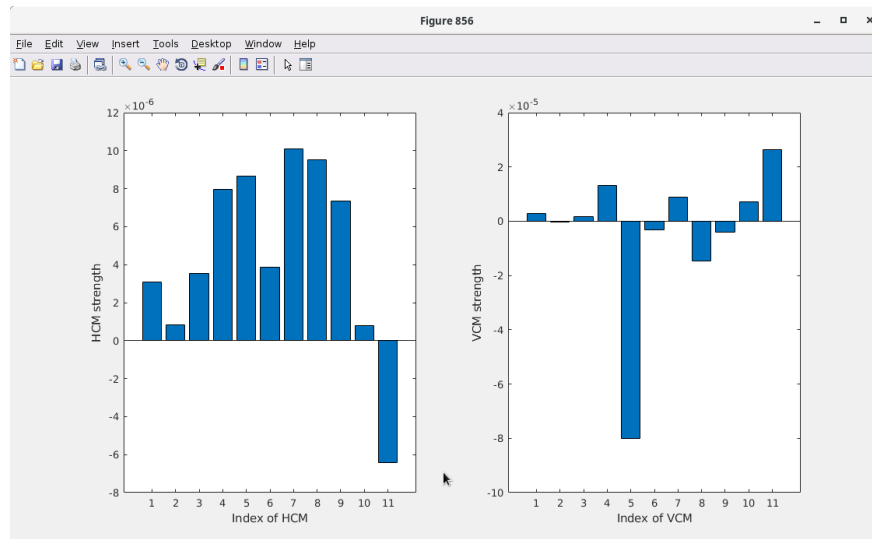
Note: You can choose to plot the corrector magnet's steps by checking the checkbox.



Step 15: Calculation of the Corrector Magnet Step

16 Plot Corrector Magnet Step

Instruction: If you chose to plot the corrector magnet step, this is how it could look.



Step 16: Visualization of the Corrector Magnet Step

17 Apply Corrector Magnet Step

Instruction: Apply the corrector magnets by pushing the **Apply CM Step** button.



Step 17: Applying the Corrector Magnet Step

18 Trajectory Correction Result

Instruction: Congratulations! The trajectory is notably corrected.

Note: You can repeat the trajectory correction by calculating a new CM step and then applying it.



Step 18: Result: Corrected Trajectory